A testing-effort artefact in the most-sensitive aquatic endpoint: structure-based PFAS screening recovers predictive validity but still cannot support clearance

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Abstract

Regulatory hazard screening commonly summarises a substance by the most sensitive effect concentration reported for it. We show that this convention makes the response variable a function of how much the substance has been tested, and that the resulting artefact is large enough to invert the apparent performance of structure-based models. Using the full U.S. EPA ECOTOX Knowledgebase, we assembled 106,095 freshwater acute records for 4,757 chemicals and 15,991 freshwater records for 103 per- and polyfluoroalkyl substances (PFAS). Across PFAS, the most-sensitive minimum correlated with the logarithm of the number of available records at Spearman $\rho = -0.73$ ($p = 1 \times 10^{-18}$): substances with more than 300 records had a median protective value 6.4 $\log_{10}$ units below those with three or fewer. Subsampling every substance to equal depth halved the association ($\rho \approx -0.29$), identifying roughly half of it as a pure extreme-order- statistic artefact and the remainder as genuine reverse causation. The same pattern held across all 4,757 chemicals, where the acute minimum carried three times the confound of the acute median ($\rho = -0.46$ versus -0.16). Holding a published random-forest model fixed and changing only the external target, rank and point prediction tracked the confound: the minimum gave $R^2 = -0.75$ and $\rho = -0.02$, whereas an acute median, whose effort association is negligible ($\rho = -0.04$), gave $R^2 = +0.22$ and $\rho = +0.42$. Rebuilding training and

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external sets consistently on the acute median raised broad cross-validation to $R^2 = 0.44 \pm 0.03$, restored external validity on 58 structure-locked PFAS ($R^2 = +0.37$; $\rho = +0.46$, 95% CI $[+0.22, +0.65]$), and halved the PFAS-conditional 90% interval from 10.53 to 4.46 $\log_{10}$ mg/L. Clearance nevertheless failed: because the acute median is 2.69 $\log_{10}$ units less protective than the minimum on the same substances, a threshold matched for stringency cleared none of 16,070 in-domain inventory structures, and the Globally Harmonised System acute Category 1 boundary cleared 1.9%. The reported inability of structure-based models to clear untested PFAS is therefore correct but was attributed to the wrong cause: the models can predict this endpoint once it is constructed without an effort confound, and what still prevents clearance is interval width relative to the decision threshold.

Keywords: PFAS, aquatic ecotoxicity, hazard screening, data curation, conformal prediction, ECOTOX

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1. Introduction

Screening programmes for per- and polyfluoroalkyl substances (PFAS) face an inventory far larger than the measured record. The U.S. EPA CompTox Chemicals Dashboard lists tens of thousands of structures [1, 2], while measured freshwater effect concentrations exist for a few hundred. Structure-based models are the obvious way to triage that gap, and quantitative structure–activity relationship (QSAR) models for aquatic toxicity have a long history [3, 4] with current environmental applications combining curated descriptors, fingerprints and machine learning [5, 6]. Best-practice guidance emphasises chemically independent validation, transparent applicability domains and an explicit endpoint definition [7, 8].

That a model trained on conventional organic pollutants transfers poorly to PFAS is established [9], and the usual explanation is chemical: perfluorinated chemistry is absent from the training distribution, so the model is extrapolating. We report evidence that a second mechanism, independent of chemistry, contributes substantially to the observed failure and has not to our knowledge been documented.

The mechanism concerns how the response variable is built. Regulatory hazard assessment is conventionally protective: a substance is summarised by the lowest effect concentration reported for it, across species, exposure durations and effect types. The practice is sensible as policy, because the

most sensitive observed effect bounds the hazard. As a statistical construction, however, the minimum of a sample is an extreme order statistic, and its expected value decreases as the sample grows. Two substances with identical toxicity will therefore receive different protective values if one has been tested more often than the other. Because testing effort varies by orders of magnitude across the ecotoxicity record — a handful of records for most substances, thousands for PFOA and PFOS — this is not a second-order concern.

If that reasoning is right, three things should follow, and all three are testable. The protective minimum should correlate with the number of available records; equalising the number of records should attenuate the correlation; and a structure-based model, which has no access to testing history, should fail on the protective minimum while succeeding on an endpoint constructed without the confound. We test each in turn, using the complete ECOTOX Knowledgebase rather than a curated subset, and then ask what the corrected picture implies for the decision the screening programme actually wants to make: whether an untested PFAS can be set aside as low concern.

2. Materials and methods

2.1. Data source and record selection

All toxicity data came from the U.S. EPA ECOTOX Knowledgebase ASCII release of 15 September 2026 [10], used in full rather than through a curated derivative. From 725,636 test records we retained tests on aquatic organisms (`organism_habitat` = Water) in freshwater media (`media_type` beginning FW), joined them to 1,250,612 result records, and kept results reporting an acute LC50, EC50 or IC50 or a chronic NOEC, LOEC or MATC in mass-concentration units convertible to mg/L. Records carrying an inequality qualifier were identified and excluded from the primary analysis rather than treated as point observations, because left-censored values have maximum leverage on the extreme order statistics examined here. This yielded 106,095 acute freshwater records for 4,757 chemicals, and, for the PFAS subset defined below, 15,991 acute and chronic records for 103 substances.

PFAS were identified by CAS match to the PFASSTRUCTV6 list of the CompTox Chemicals Dashboard [2], the same operational definition used in the inventory we later deploy against. Structures were resolved from CAS

through PubChem; 4,311 of 4,757 chemicals (90.6%) resolved successfully, and unresolved chemicals were dropped.

2.2. Endpoint constructions

Four summaries of the same records were compared. The *minimum* is the lowest reported value across all acute and chronic strata, which is the protective convention. The *median* is the median over the same records. The *acute median* restricts to acute lethality and concentration-response endpoints before taking the median. The *depth-equalised minimum* draws a fixed number k of records per substance without replacement and averages the resulting minima over repeated draws, isolating the effect of sampling depth while holding the number of records constant by construction. All are expressed as $y = \log_{10}(T/\mathrm{mg\,L^{-1}})$.

Testing effort was quantified as $\log_{10}$ of the number of usable records for a substance, and its association with each endpoint was measured by Spearman correlation. We emphasise that this is a property of the response variable alone; no model is involved.

2.3. Structure standardisation

Structures were desalted to the fluorine-bearing fragment where present, charge-neutralised with the RDKit standardiser and canonicalised [11, 12]. Desalting to the largest fragment, the more common convention, misassigns salts whose counterion exceeds the active moiety in size, and we therefore prefer the fluorinated fragment explicitly. Substances collapsing to the same parent structure were represented once, using the median of their values on the chosen endpoint. PFAS structures and inventory structures were removed from the broad training set before fitting.

2.4. Model, validation and conformal intervals

The representation was a 2,048-bit ECFP4 fingerprint (Morgan radius 2) [13] and the learner a random-forest regressor with 300 trees, `max_features` $= \sqrt{p}$ and seed 20260620 [14, 15]; these are the settings of the published analysis we re-examine, held fixed so that differences are attributable to the endpoint and not to tuning. Broad performance used five repeats of shuffled five-fold cross-validation over structure-locked rows. External validation fitted the full broad model once and evaluated it on the PFAS set, with substance-level bootstrap confidence intervals (4,000 resamples) on Spearman ρ.

Normalised split-conformal intervals used the nonconformity score

$$s_i = \frac{|y_i - \hat{y}_i|}{\sigma_i + \beta}, \tag{1}$$

with σ_i the standard deviation across forest trees and β the median calibration σ plus 10^{-6}, and the finite-sample order statistic $\lceil (n+1)(1-\alpha) \rceil$. Marginal calibration used a random 80/20 split of the broad set, evaluated on the PFAS set. PFAS-conditional calibration held the broad model fixed and split the PFAS set into disjoint calibration and test halves. Both were repeated 40 times and means reported.

2.5. Applicability domain and clearance

The applicability domain was the mean Euclidean distance in ECFP4 space to the five nearest training structures, thresholded at the 95th percentile of training leave-self-out distances. A structure was *cleared* when the lower bound of its 90% interval exceeded the hazard threshold. Because the endpoint constructions differ in protectiveness, clearance was evaluated across a continuum of thresholds rather than at a single value, and reported at three reference points: 0.1 mg/L, the value used with the protective minimum; 1 mg/L, the Globally Harmonised System acute aquatic Category 1 boundary; and a stringency-matched value obtained by shifting 0.1 mg/L by the measured offset between the two endpoints.

2.6. Subclass assignment

PFAS were assigned to structure categories following the OECD (2021) definition and the 2018 OECD/UNEP structure series [16, 17]. The definitional gate requires at least one fully fluorinated, saturated methyl or methylene carbon with no hydrogen, chlorine, bromine or iodine attached; methine (>CF–) and unsaturated (=CF_2) carbons do not satisfy it. Categories were then assigned by parsing the molecule as C_nF_{2n+1}–X–R, with head-group identity taking precedence over incidental fluorination, so that a perfluoroalkyl ether acid is recorded as an ether acid rather than as a chain compound.

3. Results and discussion

3.1. The protective minimum tracks how much a substance has been tested

Across the 103 PFAS with freshwater records, the protective minimum correlated with the logarithm of the number of available records at $\rho = -0.73$

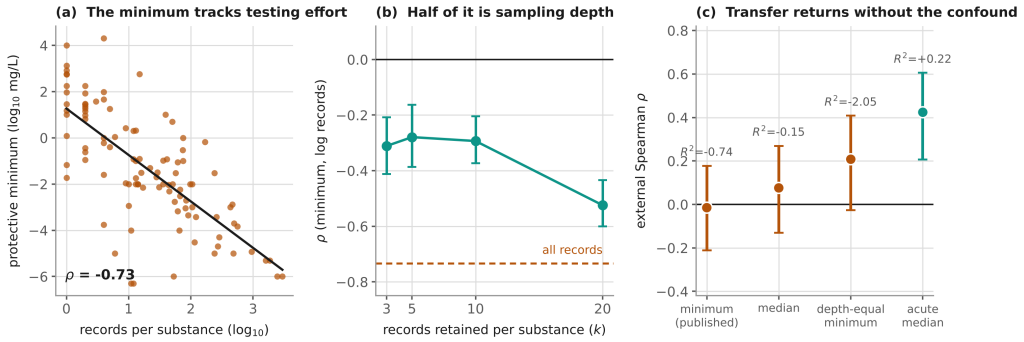


Figure 1: The protective endpoint is confounded with testing effort. (a) Protective minimum against the number of freshwater records per PFAS substance ($n = 103$). (b) Spearman correlation between the minimum and the record count after subsampling every eligible substance to the same depth k; the dashed line is the value obtained using all records, and bars are 2.5th–97.5th percentiles over 200 draws. (c) External performance of a fixed broad non-PFAS model evaluated against four constructions of the same underlying records, with 95% bootstrap intervals on Spearman ρ and R^2 annotated; teal marks an interval excluding zero.

($p = 1 \times 10^{-18}$; Figure 1a). The gradient is not subtle. Substances with three or fewer records had a median protective value of +1.40 $\log_{10}$ mg/L; those with more than 300 records had a median of −4.96. The difference of 6.4 $\log_{10}$ units between the least- and most-studied strata is larger than the entire range the modelling literature attempts to predict.

The same substances illustrate the mechanism individually. Perfluorooctanoic acid carries 2,966 freshwater records spanning 9.03 $\log_{10}$ units, from −6.00 to +3.03; potassium perfluorooctanesulfonate carries 2,434 spanning 8.60. Under the protective convention each is summarised by the lowest value in its own large sample. These extreme values are not data-entry errors — they are genuine entries in a public, curated database — but they are the minima of very large draws from very wide distributions, and a substance tested three times cannot produce a comparable minimum however toxic it is.

3.2. About half of the association is a sampling-depth artefact

Correlation between effort and the protective value admits two explanations that are not mutually exclusive. Substances of greater concern attract more testing, which would make the association causal in the reverse direction; and the minimum of a larger sample is lower by construction, which

would make part of it an artefact. Subsampling separates them. Drawing the same number of records from every eligible substance and recomputing the minimum reduced the correlation from -0.73 to -0.31 at $k = 3$, -0.28 at $k = 5$ and -0.29 at $k = 10$ (Figure 1b). Roughly half the association therefore disappears when sampling depth is held constant, and roughly half remains as genuine reverse causation. The value at $k = 20$ (-0.52) is higher but rests on the 47 substances with at least 20 records, which are themselves the well-studied ones, and we do not read it as contradicting the pattern.

The effect is not peculiar to PFAS. Over all 4,757 chemicals with freshwater acute data, the acute minimum carried $\rho = -0.46$ against record count while the acute median carried $\rho = -0.16$ — a threefold difference between two summaries of identical records. The confound is a property of taking a minimum, not of the chemical class.

3.3. Transfer failure follows the confound, not the chemistry

A structure-based model has no access to a substance's testing history. If the protective endpoint is partly a measure of that history, the model should fail on it for reasons unrelated to chemistry. Holding the published broad non-PFAS model fixed and changing only the external target tests this directly (Figure 1c). On the protective minimum the model gave $R^2 = -0.745$ and $\rho = -0.015$ (95% CI $[-0.21, +0.18]$). On the median of the same records it gave $R^2 = -0.147$ and $\rho = +0.076$. On an acute median, whose association with record count is $\rho = -0.04$, it gave $R^2 = +0.216$ and $\rho = +0.424$ (95% CI $[+0.21, +0.61]$), the only configuration whose interval excludes zero.

Across the four constructions, the ordering of external performance is the ordering of freedom from the effort confound. We regard this correspondence as the central evidence of the paper: the same model, the same molecules and the same underlying measurements yield a failed or a successful external validation depending only on how the response variable was summarised.

This also accounts for an observation reported previously without a mechanism. Restricting to acute endpoints has been noted to improve external rank correlation, and attributed to heterogeneity in assay availability. The present analysis identifies what that heterogeneity is: acute-only restriction removes most of the effort confound, because acute lethality endpoints are reported in a narrower range and the median is not an extreme order statistic.

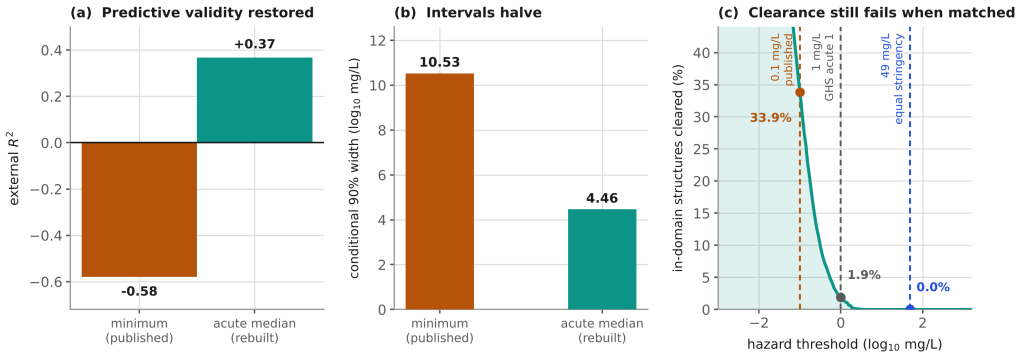


Figure 2: Consistent rebuilding on the acute median. (a) External R^2 on structure-locked PFAS for the published protective endpoint and the rebuilt acute-median endpoint. (b) Median PFAS-conditional 90% interval width. (c) Fraction of in-domain inventory structures cleared as a function of the hazard threshold, with three reference thresholds marked: the value used with the protective endpoint, the GHS acute Category 1 boundary, and a value matched to the protective endpoint for stringency.

3.4. Consistent rebuilding restores predictive validity

The comparison above changed the external target while leaving the training target as published. Rebuilding both consistently on the acute median gave a broad training set of 3,718 structure-locked non-PFAS chemicals and an external set of 58 structure-locked PFAS. Broad cross-validation improved from $R^2 = 0.387 \pm 0.024$ to 0.443 ± 0.029. External validation improved from $R^2 = -0.580$ to $+0.367$ and from $\rho = -0.071$ to $+0.460$ (95% CI $[+0.22, +0.65]$) (Figure 2a).

Interval behaviour improved correspondingly. Marginal calibration, which under-covered PFAS badly on the protective endpoint (0.659 at nominal 0.90), reached 0.858. PFAS-conditional calibration reached 0.897, and its median 90% interval narrowed from 10.53 to 4.46 $\log_{10}$ mg/L (Figure 2b). An interval spanning more than ten orders of magnitude covers the whole range from ambient background to the most heavily impacted discharge concentrations on record; one spanning four and a half does not, and is for the first time of a width at which an environmental decision could in principle be made.

3.5. Clearance still fails once the threshold is matched

Applied to the untested inventory, the rebuilt model placed 16,070 of 17,619 unique structures inside the applicability domain with a median in-

terval width of 4.87 $\log_{10}$ units, and cleared 5,441 of them (33.9%) at the 0.1 mg/L threshold used with the protective endpoint. That number must not be compared with the zero reported for the protective endpoint, and the reason is quantitative. On the same substances the acute median lies 2.69 $\log_{10}$ units above the protective minimum, and 94.3% of measured PFAS exceed 0.1 mg/L on the acute median against 42.9% on the minimum. The threshold means something different on each endpoint, and holding it fixed while changing the endpoint relaxes the decision rather than improving it.

Evaluating clearance across thresholds removes the ambiguity (Figure 2c). At the GHS acute aquatic Category 1 boundary of 1 mg/L, 306 structures clear (1.9%). At a threshold shifted by the measured 2.69-unit offset, which matches the protective endpoint for stringency, none clear. The published conclusion that no untested PFAS can be cleared therefore survives, and survives on a model that now has demonstrated external validity rather than on one that had none.

What changes is the explanation, and the two reference thresholds fail for different reasons. The obstacle is not that structure-based models cannot predict PFAS aquatic toxicity; on a properly constructed endpoint they predict it with $R^2 = +0.37$ and a rank correlation whose interval excludes zero. At the GHS Category 1 boundary the obstacle is uncertainty: the median in-domain structure would clear with a 90% interval of 2.43 $\log_{10}$ units against the 4.87 achieved, so halving the interval again would be sufficient. That is a concrete target, and one the previous framing could not have specified, because an endpoint confounded with testing effort does not define a stable distance to any threshold.

At the stringency-matched threshold the obstacle is different and more fundamental. There the median in-domain structure has a point prediction *below* the threshold, so the required half-width is negative and no narrowing of the interval would clear it. At protective stringency the model is not uncertain about whether these structures are low hazard; it predicts that most of them are not. Separating these two regimes matters for research planning, because only the first is addressed by better uncertainty quantification.

3.6. Structure classes cannot be resolved at available sample sizes

A natural next question is whether transfer differs across PFAS structure classes, since the chemical argument for extrapolation failure is class-specific. We tested this on OECD (2021) categories and report a negative result with its power requirement, because the point estimates invite over-reading.

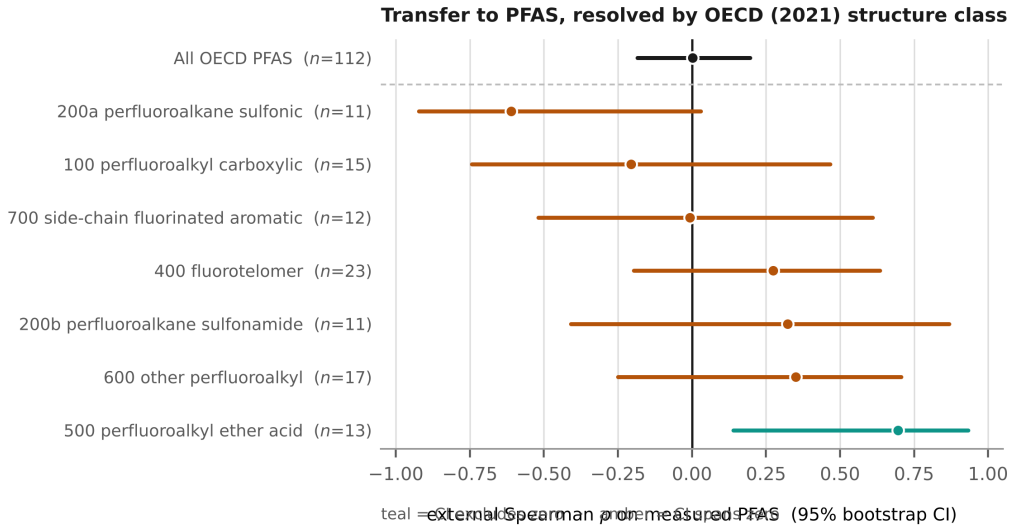


Figure 3: External rank correlation by OECD (2021) structure class on the protective endpoint, with 95% bootstrap intervals. Only perfluoroalkyl ether acids exclude zero, on 13 substances and as one of seven comparisons.

On the protective endpoint, where the measured set is largest ($n = 112$ after applying the OECD definitional gate, which excludes seven substances including difluoroacetic acid, whose CHF_2 carbon is not fully fluorinated), only one of seven categories had a bootstrap interval excluding zero: perfluoroalkyl ether acids at $\rho = +0.70$ (95% CI $[+0.14, +0.93]$, $n = 13$; Figure 3). Every other category, including the perfluoroalkyl acids on which extrapolation is expected to be worst, spanned zero. The single positive result sits close to the detection floor for its sample size and is one of seven comparisons, so we do not interpret it as established.

The acute-median set is smaller still. Of 68 OECD-classified PFAS, three categories reach ten substances and the largest reaches fourteen. At those sizes, 80% power at $\alpha = 0.05$ detects only $|\rho| \geq 0.69$. Resolving differential transfer at a more realistic effect size of $\rho = 0.3$ would require approximately 85 measured substances per category, or 133 with correction for seven comparisons — roughly 600 to 900 measured PFAS against the 103 the public record currently supports. This is the quantity a programme that wants class-resolved screening would need to measure, and it is five to nine times what exists.

3.7. Implications for screening practice

Three consequences follow for programmes that build or consume structure-based hazard screens.

First, the response variable deserves the scrutiny usually reserved for the model. A protective minimum taken over a heterogeneous record is not a property of the substance alone, and the degree to which it is not can be measured directly with the correlation reported here. We would suggest that any screening study report the association between its endpoint and the number of underlying records, as a routine diagnostic, in the way applicability domains are now routinely reported.

Second, a model that fails external validation on a protective endpoint has not thereby been shown to lack chemical validity. The failure is consistent with the model being unable to predict testing history, which is not a chemical deficiency and is not repaired by a better representation or a larger training set. Distinguishing the two requires exactly the endpoint comparison performed here, and it is inexpensive.

Third, the protective convention is not thereby wrong. It remains the correct posture for a regulatory decision, because the lowest observed effect does bound the hazard. What the present results argue is that it is unsuitable as a *regression target*, and that screening pipelines should model an effort-stable quantity and apply protectiveness at the decision step — through the threshold and the assessment factor — rather than embedding it in the response variable where it cannot be separated from testing history.

3.8. Limitations

The external PFAS set contains 58 structure-locked substances on the acute median, smaller than the 119 available on the protective endpoint, because restricting to acute endpoints and collapsing salts both reduce it. This is the principal limitation and it bounds the precision of every external statistic reported. Structures resolved for 90.6% of chemicals; the unresolved remainder may be non-random with respect to chemical class, which we could not assess. The acute median remains a mixture over species, exposure duration and effect type, and is effort-stable rather than effort-free. The clearance analysis evaluates aquatic effect concentrations only and says nothing about exposure, persistence or bioaccumulation, which are the properties that in practice drive PFAS regulatory action; a substance with a high acute LC50 and a decades-long half-life is not low concern, and no aquatic-toxicity model

of any quality would establish that it was. Finally, the reverse-causation component of the effort–toxicity association — roughly half of it — is real and is not removed by changing the endpoint summary.

4. Conclusions

Summarising a substance by the lowest effect concentration ever reported for it makes the resulting value depend on how often it has been tested. Across 103 PFAS the protective minimum correlated with record count at $\rho = -0.73$, and subsampling to equal depth identified about half of that as an extreme-order-statistic artefact; across 4,757 chemicals the acute minimum carried three times the confound of the acute median. Because a structure-based model cannot predict testing history, this confound is sufficient on its own to produce the external validation failures reported for PFAS screening: holding a published model fixed, the same molecules gave $R^2 = -0.75$ on the protective minimum and $R^2 = +0.22$ on an effort-stable acute median.

Rebuilding training and external sets consistently on the acute median restored external validity ($R^2 = +0.37$; $\rho = +0.46$, 95% CI $[+0.22, +0.65]$) and halved the PFAS-conditional interval to 4.46 $\log_{10}$ mg/L. Clearance nevertheless failed at a stringency-matched threshold, where no in-domain inventory structure cleared, and at the GHS acute Category 1 boundary, where 1.9% did. The conclusion that structure-based screening cannot presently clear untested PFAS therefore stands, but it stands for a reason the earlier framing did not identify: not that such models cannot predict PFAS aquatic toxicity, but that a validated model's interval, and at protective stringency its point prediction, remain on the wrong side of the decision threshold. The practical recommendation is to model an effort-stable endpoint, report the endpoint–effort association as a routine diagnostic, and apply protectiveness at the decision step rather than inside the response variable.

CRediT authorship contribution statement

Jeongyun Kim: Methodology, Software, Formal analysis, Writing – original draft. **Jin Yoo:** Data curation, Validation, Formal analysis, Writing – review & editing. **Ku Kang:** Conceptualization, Supervision, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

During the preparation of this work the authors used a large language model to assist with code development for data extraction and analysis, and with language editing. The authors reviewed and edited all output and take full responsibility for the content of the publication.

Data availability

All extraction and analysis scripts, the structure-locked tables, and the inventory-level predictions are openly available at `https://github.com/bisu9082/pfas-clearance-validation`, tagged as release `v3.0`. The underlying source data are public: the U.S. EPA ECOTOX Knowledgebase ASCII release of 15 September 2026, the CompTox Chemicals Dashboard PFASSTRUCTV6 list, and PubChem for structure resolution. Raw third-party databases are not redistributed; the repository documents the exact release used and the processing pipeline that regenerates every reported number.

Acknowledgements

The authors thank the teams maintaining the U.S. EPA ECOTOX Knowledgebase, the CompTox Chemicals Dashboard and PubChem.

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